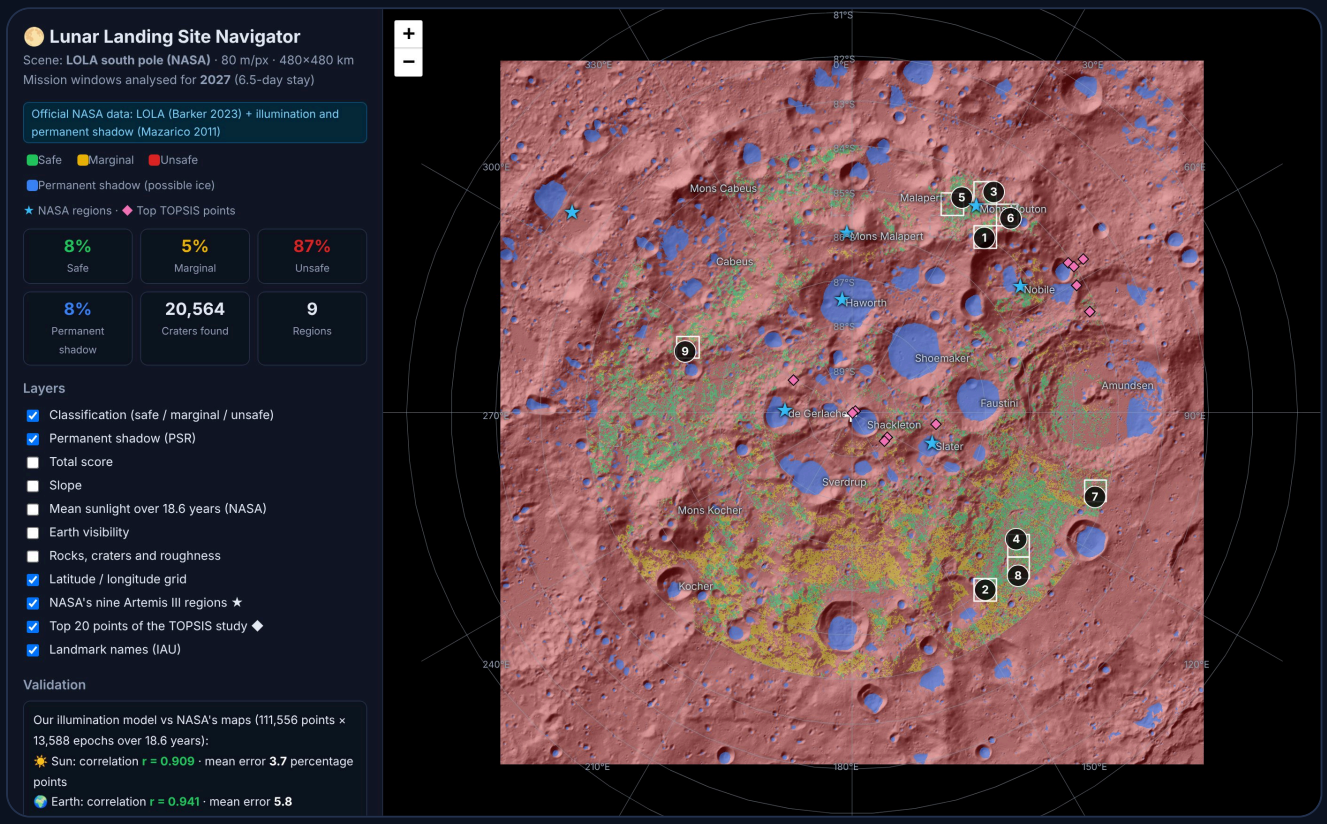
LUNAR LANDING SITE NAVIGATOR · TECHNICAL REPORT

Lunar Landing Site Navigator

The complete report, from A to Z

An open-source system that picks the best landing sites at the lunar south pole from official NASA data and the real 2027 launch windows. Its accuracy is checked against NASA's own maps, an independent published study, and NASA's official Artemis III regions.



NASA PGDA data · LOLA · Barker 2023 · Mazarico 2011 · Python · numba · astropy · GitHub Actions

Version 1.0 · November 2026 · github.com/ELD11q/-ahs · [Team name]

Executive summary

The idea and the result on one page

The problem: the lunar south pole is the first target of Artemis III because it has craters in permanent shadow that may hold water ice. But the Sun never rises more than about 1.5° above the horizon there, so shadows are long and the terrain is rough. Choosing where to land means analysing a very large amount of data.

The solution: a Python system that downloads 6 official NASA products (1.2 GB) and puts them on one 480×480 km grid at 80 m (36 million pixels). It then scores every point on five criteria: slope, sunlight, Earth visibility, distance to ice, and rocks. It picks 13 candidate regions, then the best 9, and computes the real 2027 launch windows for each one, hour by hour.

0.909	0.941	70%	4/9
correlation of our Sun model with NASA's maps	correlation of our Earth-visibility model with NASA	of the TOPSIS study's best points fall in our top 10%	of NASA's Artemis III regions covered by our regions

Best region: 85.08°S 37.2°E near Mons Mouton. It gets 6.5 days of continuous sunlight for **36%** of the possible 2027 start times, Earth is visible 81% of the time, and the nearest permanent shadow (possible ice) is 2.33 km away. The best launch window starts on 2027-01-09. It stays the best region in 2026, 2027 and 2028.

The value: work that takes weeks of manual GIS analysis comes out of one command in about 3 minutes, and it can be repeated for any year or set of constraints. The core is open source; the Open-Core business model is in chapter 10.

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1 The problem, and why the south pole

The most important place on the Moon, and the hardest place to land

Why it matters

- Craters in **permanent shadow**, at about −230 °C, that may hold water ice.
- Water means drinking water, oxygen, and hydrogen fuel for rockets.
- Nearby peaks are **lit most of the time**, which means almost continuous solar power.

Why it is hard

- The Sun never rises more than ~1.5° above the horizon, so any hill blocks it.
- Earth sometimes drops below the horizon and direct contact is lost.
- Slopes, craters and boulders; the lander needs a slope under 8° to 10°.

What makes a site safe

Condition	Value in the system	Reason
Slope	red above 10°, yellow between 8° and 10°	keep the lander stable and upright
Sunlight	6.5 days of continuous light (the mission length)	power and temperature
Contact with Earth	share of time Earth is above the horizon	direct-to-Earth (DTE) communication
Near permanent shadow	ideal ≤ 2 km, no value beyond 15 km	walking to the ice
Rocks and craters	roughness at 100 m + detected craters	landing safety
Exploration zone	within 6° of the pole (~84°S)	Artemis III requirement
Landing ellipse	a 200 m radius circle around each point must be fully safe	limited landing accuracy

2 How NASA chooses landing sites

The LRO spacecraft, and from 13 regions to 9

The **LRO** spacecraft (Lunar Reconnaissance Orbiter) has orbited the Moon since 2009 and collects data with three instruments:

Instrument	Measures	In our project
LOLA	laser heights, and from them slope, roughness and illumination	Barker 2023 maps + Mazarico 2011 illumination maps
Diviner	temperature, and from it rock abundance	roughness from LOLA, with a `--rocks` option
LROC NAC	~0.5 m images of craters and boulders	ready YOLO module (`hazards.py`)

In 2022 NASA announced **13 candidate regions** for Artemis III, each about 15×15 km. In October 2024 it narrowed them to **9 regions**: Peak near Cabeus B, Haworth, Malapert Massif, Mons Mouton Plateau, Mons Mouton, Nobile Rim 1, Nobile Rim 2, de Gerlache Rim 2 and Slater Plain. According to NASA's summary (NTRS 20250011660), the downselection was based on:

- Year-round *mission availability*: all launch, orbit and lighting constraints combined.
- Terrain safety: a "sufficient" low-slope area, avoiding craters and boulders.
- Direct contact with Earth, and continuous lighting for the whole surface stay (5.75 to 6.25 days).
- Keeping several regions, because "each region alone offers limited opportunities".

> Our project follows the same logic: we pick 13 and then 9, give 40% of the final ranking to launch-window availability, reward a "sufficient" safe area rather than the "largest", and force a distance between regions so they are diverse.

3 System architecture

From raw files to a landing decision

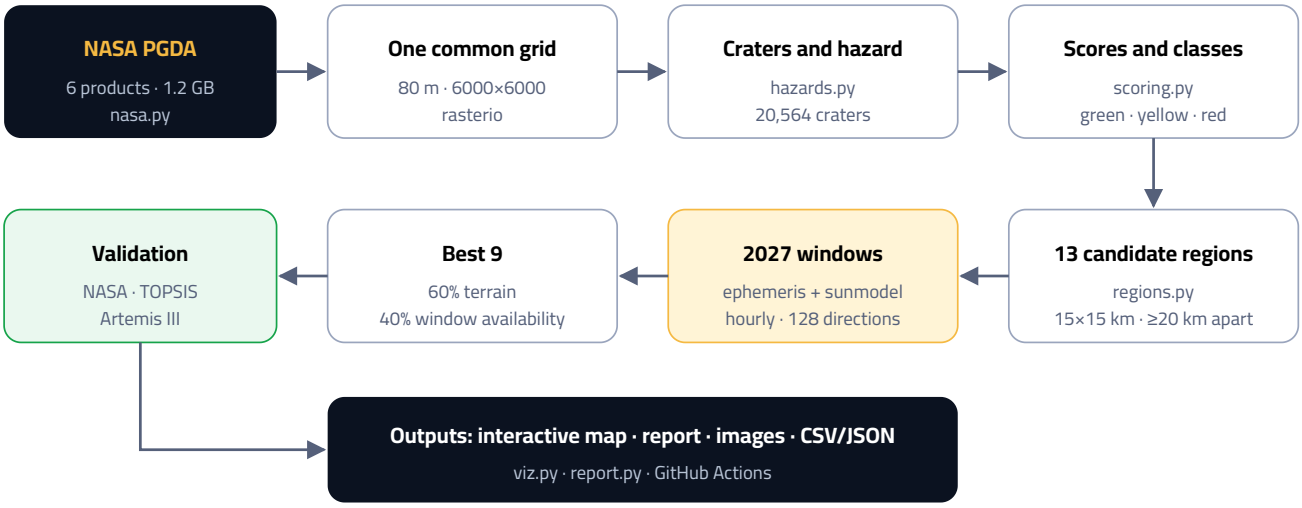


Figure 1: the stages of the system; every box is a separate, tested module

Code modules

File	Role
`nasa.py`	product catalogue, resumable download, reprojection onto one grid
`ephemeris.py`	Sun and Earth positions (astropy/ERFA) + lunar orientation (IAU 2009)
`sunmodel.py`	horizon ray-tracing and time-series visibility (numba, parallel)
`terrain.py` · `hazards.py`	slope, roughness, crater detection, optional YOLO, hazard map
`scoring.py` · `regions.py`	weighted score, classes, region selection, coordinate conversion
`nasa_pipeline.py`	ties everything together: candidates, windows, final pick, validation
`reference.py`	NASA's nine regions, the TOPSIS top 20 points, IAU landmarks
`experiments.py`	robustness tests: weights, slope limit, years
`viz.py` · `report.py` · `cli.py`	interactive map, images, report, command line

4 The official data

6 products from NASA Goddard, free and open

Layer	File	Resolution	Reference
Elevation	LDEM_80S_80MPP_ADJ.TIF	80 m	Barker et al. 2023 · PGDA 90
Slope	LDSM_80S_80MPP_ADJ.TIF	80 m	Barker et al. 2023
Roughness (100 m)	LDRM_80S_50MPP_ADJ_ROUGH_100M.TIF	50 m	Barker et al. 2023
Average illumination	AVGVISIB_75S_120M_201608.TIF	120 m	Mazarico et al. 2011 · PGDA 69
Average Earth visibility	AVGVISIB_75S_120M_201608_EARTH.TIF	120 m	Mazarico et al. 2011
Permanent shadow	LPSR_75S_120M_201608.TIF	120 m	Mazarico et al. 2011

- All files use a polar stereographic projection (sphere of radius 1737.4 km, longitude 0 pointing up), so they line up exactly with no distortion.
- We checked the coordinates: the centre of Shackleton crater (from the IAU gazetteer) comes out at −2788 m depth inside a permanent shadow, and the inverse transform returns the same coordinates.
- The illumination averages are computed hour by hour over a full 18.6-year cycle and include the polar winter. That is why even the best sites average only about 0.35 to 0.5, and we calibrated the thresholds on this distribution: median 0.29, top 1% at 0.49.

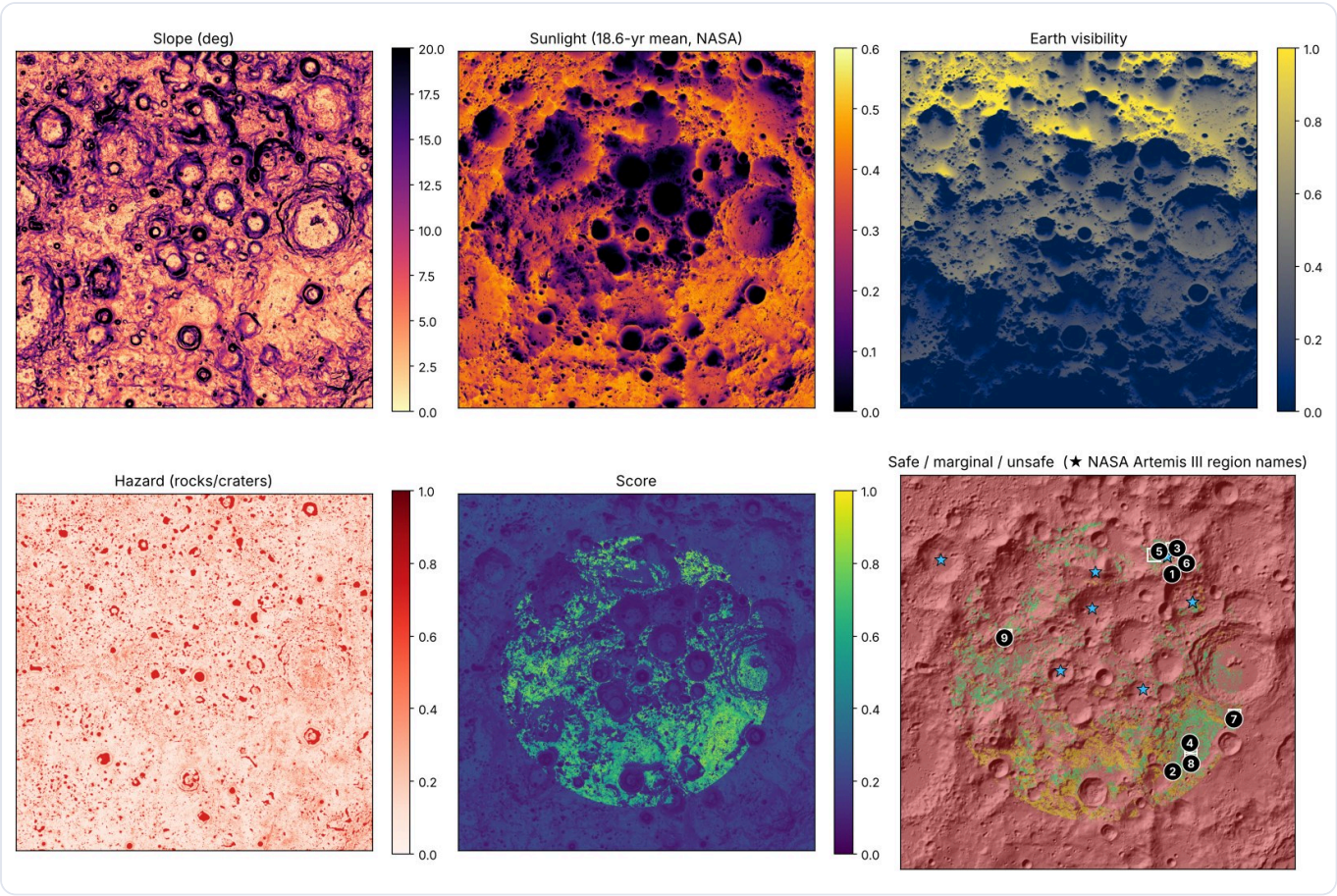


Figure 2: the layers on real data: slope, average illumination, Earth visibility, hazard, score, and classes inside the exploration zone (stars = NASA regions)

5 Method, step by step

How every pixel becomes a decision

5.1 Slope, roughness and craters

Slope comes from NASA's official product. Roughness at a 100 m baseline becomes a 0 to 1 hazard index (above 3 m counts as rocky and rough). Craters are detected from the elevation map by subtracting a smoothed copy at 3 scales, which finds 20,564 craters larger than 400 m.

5.2 The physical illumination engine

1. **Horizon:** for each point we trace rays over the LOLA map in 64 to 128 directions out to 150 km, with bilinear sampling and the Moon's curvature $d^2/2R$.
2. **Sun and Earth positions:** from the ERFA ephemeris built into astropy, converted to the lunar frame with the IAU 2009 model (Archinal 2011). Checked values: Sun elevation ±1.55°, Earth libration ±6.8° and ±7.7°, distance 357 to 407 thousand km.
3. **Visibility:** for every moment we compute the body's elevation and azimuth in the grid frame (with parallax for Earth), compare it with the horizon, and compute the visible part of the disc (Sun 0.53°, Earth 1.9°).
4. **Speed:** written in numba and runs on all cores: 111 thousand points × 13.6 thousand moments in about 35 seconds.

5.3 Scores and classes

Criterion	Conversion to 0–1	Weight
Slope	1 – slope ÷ 10°	30%
Sunlight	NASA average ÷ 0.5 (capped at 1)	25%
Earth visibility	NASA average ÷ 0.6 (capped at 1)	15%
Near permanent shadow	1 up to 2 km, then falls to 0 at 15 km	15%
Rocks and craters	1 – hazard index	15%

Red: slope above 10°, permanent shadow, illumination below 0.15, hazard above 0.6, or outside the exploration zone, all grown by 200 m for the landing ellipse. **Yellow:** slope above 8°, Earth visibility below 0.15, or score below 0.55. **Green:** everything else.

5.4 Choosing regions

A 15×15 km window slides in 7.5 km steps. Region score = 30% for the safe area (full at 15% of its area) + 70% for the mean of the top 5% of scores inside it. We then pick 13 regions that do not overlap, with at least 20 km between their

best points.

5.5 Launch windows and the final pick

For the best point of every candidate region we compute sunlight and Earth visibility every hour through the year (for an observer 2 m above the ground). A **valid window** = 6.5 continuous days with more than half of the Sun's disc visible every hour. **Availability** = the share of start times (every 6 hours) that give a valid window. **Final score** = 60% terrain score + 40% availability, and then the best 9.

6 Results

The best 9 regions on real NASA data, with 2027 windows

#	Location	Score	2027 windows	Best 2027 window (month-day)	Slope	Sunlight	Earth	Ice	Safe	NASA region
1	**85.08°S 37.2°E**	**0.992**	**36%**	**01-09**	**0.47°**	**50%**	**81%**	**2.33 km**	**33%**	**Mons Mouton / Nobile**
2	85.01°S 143.0°E	0.894	28%	03-10	0.65°	49%	33%	1.62 km	64%	—
3	84.11°S 32.7°E	0.882	26%	01-13	0.24°	49%	83%	1.19 km	43%	Mons Mouton
4	85.34°S 127.6°E	0.876	25%	03-09	0.18°	48%	33%	1.66 km	77%	—
5	84.58°S 27.0°E	0.866	24%	01-13	0.58°	48%	87%	2.17 km	51%	Mons Mouton
6	84.38°S 39.2°E	0.857	24%	01-12	0.29°	52%	80%	0.62 km	39%	Mons Mouton / Nobile
7	84.23°S 109.1°E	0.852	24%	02-10	0.29°	46%	34%	2.15 km	59%	—
8	84.76°S 134.5°E	0.843	23%	03-09	0.32°	46%	33%	0.56 km	67%	—
9	86.01°S 290.1°E	0.838	23%	06-22	0.57°	51%	56%	0.64 km	30%	—

Sunlight and Earth visibility are NASA's 18.6-year averages at the best point. "Safe" = green share of the region. "Ice" = distance to the nearest permanent shadow.

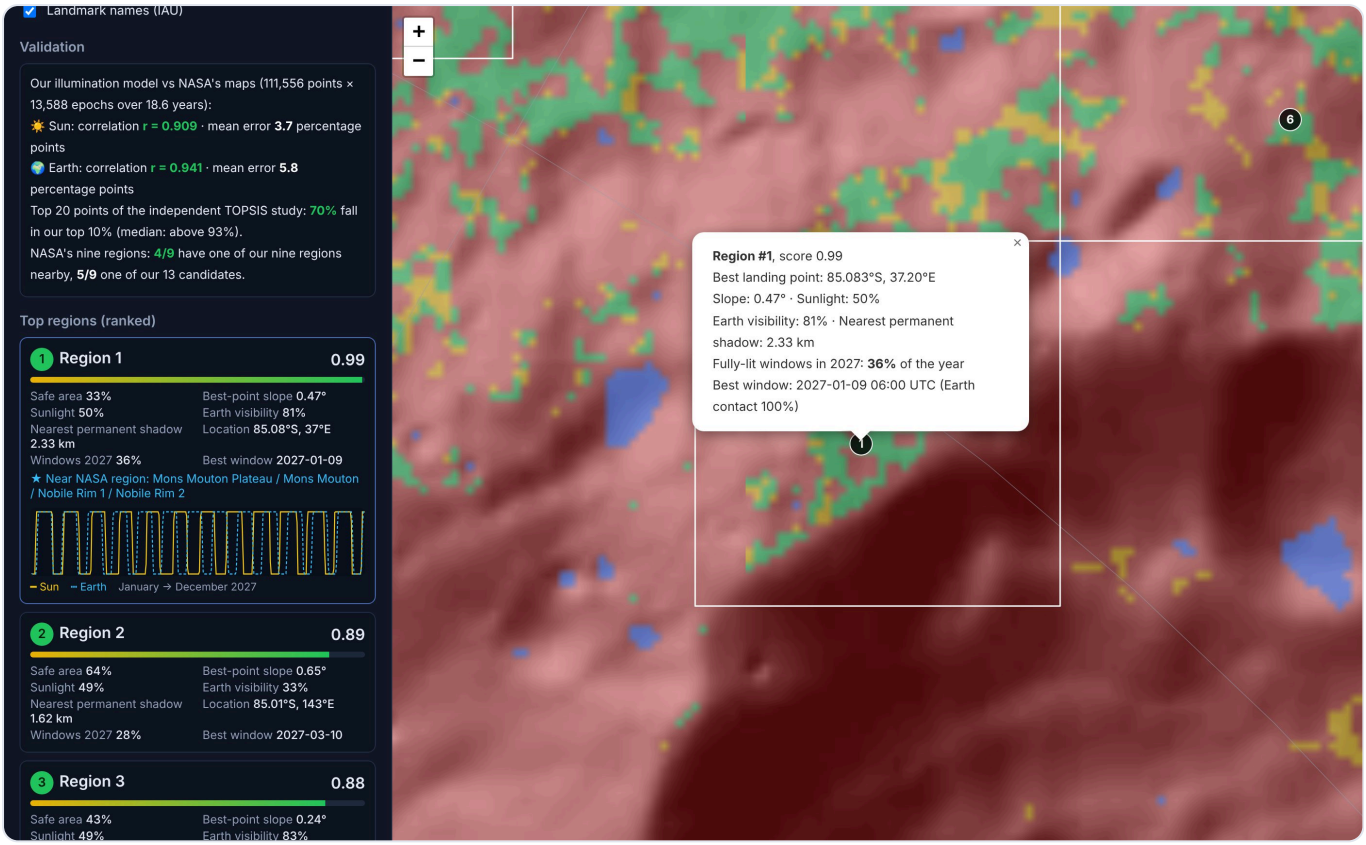


Figure 3: region 1 near Mons Mouton in the interactive map, with its details and the day-by-day Sun and Earth chart for 2027

> Four of our regions (1, 3, 5 and 6) lie around Mons Mouton, where NASA picked two regions. In every region the best point has a slope below 1°, and the nearest ice is 0.6 to 2.3 km away.

7 Accuracy checks

Three independent references

7.1 Our model vs NASA's official maps

We computed the 18.6-year average ourselves for 111,556 points × 13,588 moments and compared it point by point with the maps of Mazarico et al. (2011):

	Correlation r	Mean error	Bias	Error under 5 points	Error under 10 points
☀️ Sun	**0.909**	3.7 points	-0.6 points	78%	91%
🌍 Earth	**0.941**	5.8 points	-2.3 points	63%	84%

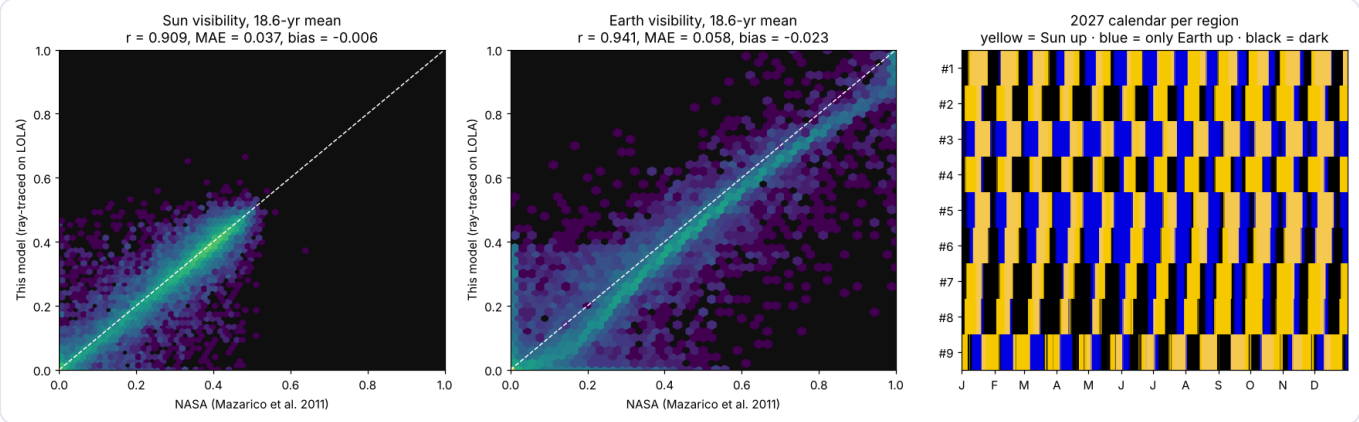


Figure 4: left and centre: our model vs NASA. Right: the 2027 calendar of each region (yellow = Sun, blue = Earth only, black = dark)

7.2 Against the independent TOPSIS study (arXiv:2406.19863)

This study ranked 1247 points in NASA's regions with TOPSIS and different data (5 m maps, horizon views, minerals). **70%** of their top 20 points fall in the top 10% of our map, and the median is above 93%. Their best point (#358, Nobile Rim 2) scores 0.763 with us, higher than 94% of the area.

Point	Region	Their R	Our score	Our percentile	Slope
358	Nobile Rim 2	0.6916	0.763	94%	2.64°
356	Nobile Rim 2	0.6869	0.752	93%	2.68°
357	Nobile Rim 2	0.6711	0.752	93%	2.68°
354	Nobile Rim 2	0.6684	0.752	93%	2.68°
350	Nobile Rim 2	0.661	0.745	92%	2.61°
355	Nobile Rim 2	0.6487	0.752	93%	2.68°
352	Nobile Rim 2	0.6455	0.785	96%	1.54°
351	Nobile Rim 2	0.6408	0.745	92%	2.61°
353	Nobile Rim 2	0.622	0.8	97%	1.33°
1121	Amundsen Rim	0.5653	0.866	100%	0.42°
1122	Amundsen Rim	0.5649	0.824	98%	1.8°
995	Nobile Rim 2	0.5353	0.11	12%	17.32°
661	Site 004 (near the pole)	0.5294	0.099	7%	30.28°
998	Nobile Rim 2	0.4916	0.866	100%	1.57°
1223	Peak Near Shackleton	0.4689	0.884	100%	0.94°
744	Faustini Rim A	0.4567	0.728	90%	1.58°
1043	Nobile Rim 2	0.4373	0.263	68%	1.16°
657	Site 004 (near the pole)	0.4179	0.096	6%	28.93°
726	de Gerlache Rim 2	0.3907	0.152	42%	20.07°
1005	Peak Near Shackleton	0.3714	0.221	67%	0.67°

7.3 Against NASA's nine regions

Our nine regions cover **4/9** of NASA's regions, and our 13 candidates cover **5/9**. NASA does not publish region boundaries, so we place each one at the centre of the landmark it is named after (IAU gazetteer), with a match radius = landmark radius + 10 km (between 20 and 50 km).

NASA region	Landmark	Radius	Our nearest region	Our regions inside	Best nearby score
Peak near Cabeus B	Cabeus B (crater)	40 km	122.7 km	—	68%
Haworth	Haworth (crater)	36 km	105.8 km	—	100%
Malapert Massif	Mons Malapert	44 km	82.0 km	—	100%
Mons Mouton Plateau	Mons Mouton	50 km	11.5 km	#1, #3, #5, #6	100%
Mons Mouton	Mons Mouton	50 km	11.5 km	#1, #3, #5, #6	100%
Nobile Rim 1	Nobile (crater)	50 km	41.3 km	#1, #6	100%
Nobile Rim 2	Nobile (crater)	50 km	41.3 km	#1, #6	100%
de Gerlache Rim 2	de Gerlache (crater)	26 km	78.6 km	—	100%
Slater Plain	Slater (crater)	23 km	87.0 km	—	100%

Why they differ: the Cabeus B landmark itself lies outside the 84° circle (while the region NASA named after it is closer to the pole). NASA also uses constraints our model does not include: the NRHO orbit, launch trajectories, the SpaceX HLS lander design, and 0.5 m LROC images. We did not tune the model to "match" NASA, so the check stays honest.

8 Robustness tests

Do the results change if we change the assumptions? (runtime 130.6 s)

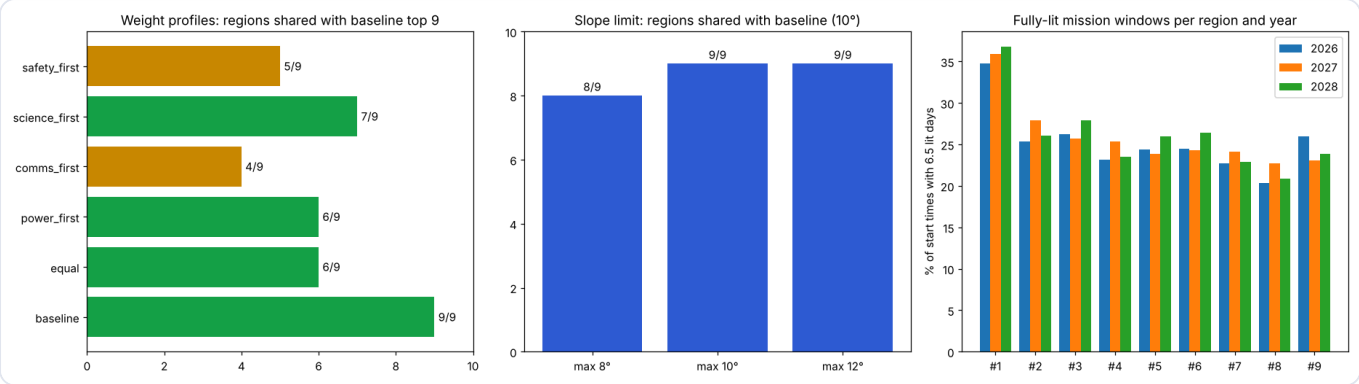


Figure 5: the effect of weights, slope limit and year on the results

8.1 Different weights for the decision maker

Profile	Slope	Sunlight	Earth	Ice	Rocks	Shared with baseline
Baseline	30%	25%	15%	15%	15%	**9/9**
Equal weights	20%	20%	20%	20%	20%	**6/9**
Power first	25%	45%	10%	10%	10%	**6/9**
Comms first	25%	20%	35%	10%	10%	**4/9**
Science (ice) first	25%	15%	10%	40%	10%	**7/9**
Safety first	45%	15%	10%	5%	25%	**5/9**

8.2 Maximum slope limit

8°	10° (baseline)	12°
8/9 (safe 6.8%)	9/9 (safe 8.2%)	9/9 (safe 8.6%)

8.3 Launch windows over three years

Region	2026	2027	2028
#1	35% 2026-01-21	36% 2027-01-09	37% 2028-01-01
#2	25% 2026-02-21	28% 2027-03-10	26% 2028-01-30
#3	26% 2026-01-01	26% 2027-01-13	28% 2028-01-02
#4	23% 2026-02-21	25% 2027-03-09	23% 2028-01-29
#5	24% 2026-01-01	24% 2027-01-13	26% 2028-01-02
#6	24% 2026-01-01	24% 2027-01-12	26% 2028-01-01
#7	23% 2026-02-21	24% 2027-02-10	23% 2028-01-02
#8	20% 2026-03-20	23% 2027-03-09	21% 2028-02-26
#9	26% 2026-01-01	23% 2027-06-22	24% 2028-05-15

Bottom line: region 1 stays the best in every year, and changing the slope limit barely changes the results. With the weights, giving top priority to Earth contact changes some regions, which gives the decision maker a clear view of the alternatives.

9 Running it from A to Z

On your computer, and on GitHub's servers

9.1 On your computer

1. Install Python 3.10 or newer, and Git.
2. Get the project:

```
git clone https://github.com/ELD11q/-ahs.git
cd -ahs/lunar-landing-navigator
python -m venv .venv && source .venv/bin/activate   # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

3. Check that everything works: `python -m pytest -q` should print 15 passed.
4. Run the full analysis on NASA data:

```
python -m lunar_nav --nasa -o output          # first run downloads ~1.2 GB
python -m lunar_nav.experiments -o output       # robustness tests (optional)
```

5. Open `output/landing_sites_map_en.html` in a browser, and read `output/REPORT_EN.md`.

9.2 Useful options

Option	What it does
`--year 2028`	launch-window year
`--max-slope 8`	slope limit
`--weights '{"illumination":0.4}'`	change the criteria weights
`--half-width-km 240 --res 80`	area size and resolution
`--demo`	fast demo mode with generated data
`--dem file.tif --pixel-size 20`	an elevation file of your choice

9.3 On GitHub's servers

The file `.github/workflows/lunar-navigator.yml` runs the tests, then the full analysis on NASA data on every update (the data is cached). To run it by hand: **Actions → Lunar Landing Navigator → Run workflow**, then pick the mode (nasa / demo / custom) and the year. The results appear on the run page and download as an artifact called **landing-sites-nasa**.

9.4 Outputs

File	Content
`landing_sites_map.html` · `landing_sites_map_en.html`	interactive map that works offline (~5 MB), Arabic and English
`REPORT.md` · `REPORT_EN.md`	report with every number, Arabic and English
`summary.png` · `validation.png`	layer and validation panels
`ranked_regions.csv/json` · `validation.json`	full data for programs and Excel

10 Business model and hackathon pitch

Open-Core: the core is open, advanced services are paid

Key partners
Saudi Space Agency, universities and research centres, a cloud provider, commercial lunar lander companies

Key activities
engine development, ongoing scientific validation, data updates, scientific publishing

Value proposition
a validated landing decision in minutes instead of weeks, with a launch-window calendar and full transparency because it is open source

Key resources
NASA open data, the physics engine, team expertise, validation results

Customer relationships
open community on GitHub, paid support, training workshops

Customer segments
space agencies, lander and payload companies, universities and robotics teams, education and hackathons

Channels
GitHub, conferences and papers, hackathons, direct partnerships

Cost structure
cloud compute and storage, team, validation, marketing. The data itself is free

Revenue streams
professional cloud subscription, custom site studies, an API, training and certificates

Proposed plans

Plan	Price	Includes	Customer
Open	free	full code, NASA data, map, report	students and researchers
Pro cloud	[$_ _ per month]	5 to 20 m resolution, custom constraints, windows for any year, API	companies and teams
Enterprise	[$_ _ per study]	full site study for a specific mission, report, support	agencies

Prices in brackets are placeholders, to be set after market validation.

Competitive edge

- NASA's internal tools are not public, and academic studies (such as TOPSIS) are one-off and done by hand in GIS software.
- Our system is **open, repeatable and validated**, and reruns the analysis for any year or constraints in minutes.
- It computes the **real launch windows**, which NASA says are decisive and which its published averages do not cover.
- Arabic and English, suited to the region's emerging space programmes.

The 3-minute hackathon pitch

1. **30 s, the problem:** the south pole has water, but landing there is hard.
2. **30 s, the solution:** one command turns NASA data into a decision.

3. **60 s, live demo:** open the map, click region 1, turn on the illumination layer.
4. **30 s, accuracy:** 0.91 and 0.94, 70%, and 4 of 9.
5. **30 s, business model and roadmap.**

Roadmap

Stage	Time	Deliverables
First model ✓	now	NASA data, 2027 windows, validation, GitHub Actions
Higher resolution	3 months	5 m LOLA maps for the final regions, LROC images + YOLO for boulders, per-pixel windows
Cloud platform	6 months	web app, API, rover path planning into permanent shadow
Expansion	12 months	the north pole, Mars, agency partnerships

11 Limits and roadmap

What the system can do, and what it cannot

- A research and education tool, **not a certified flight tool**.
- Boulders smaller than about 100 m do not show in LOLA data; they need LROC NAC images with a trained YOLO model (the code is ready, the trained model is not).
- Launch windows are computed for the best point of each candidate region, not for every pixel.
- The illumination model is very close to NASA's but not identical: the difference is 3.7 points for the Sun and 5.8 for Earth, because of horizon resolution and disc modelling.
- It does not include the NRHO orbit, launch trajectories, HLS lander constraints, or temperature.
- NASA's regions are placed at landmark centres, because their official boundaries are not published.

Code quality

- 15 automated tests, covering slope on a known surface, a reference horizon, astronomical libration, the stereographic transform, the full pipeline, and the maps (including a check that the English map has no Arabic text).
- Continuous integration on GitHub Actions on every update.
- Every number in this report is generated automatically from the result files (`docs/report/build_report.py`).

12 References and appendices

Scientific sources and data

Sources that came with the project idea

1. NASA Scientific Visualization Studio: illumination at the south pole. svs.gsfc.nasa.gov/vis/a000000/a003500/a003587/
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Data and methods

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Appendix A: glossary

Term	Meaning
PSR	Permanently Shadowed Region
LOLA / LRO	Lunar Orbiter Laser Altimeter / Lunar Reconnaissance Orbiter
DEM	digital elevation model
Libration	the Moon's apparent wobble, which moves Earth in the polar sky
TOPSIS	a multi-criteria decision method
AEZ	Artemis Exploration Zone (within 6° of the pole)
NRHO	Near-Rectilinear Halo Orbit, used by Artemis

> Repository: github.com/ELD11q/-ahs · folder `lunar-landing-navigator/` · results `results/nasa_2027/` · guide `docs/guide/` · this report `docs/report/`